جامعة خليفة
Khalifa University

Course Project

Agentic AI Based Forwarding Diagnosis in IoV Networks

COSC 630/788 Advanced Computer Networks

Final submission: Monday 22nd Nov. 2026

1 Project objective

Implement and evaluate the diagnosis system proposed in Assignment 2. Your system must analyze supplied forwarding evidence from crowdsourced Internet of Vehicles (IoV) networks and distinguish malicious forwarding behavior from benign degradation when the evidence supports that distinction. When evidence is insufficient or conflicting, the system must be able to report uncertainty rather than make an unsupported accusation.

The project investigates how agentic AI can support evidence based diagnosis. Packet loss alone is not proof of malicious behavior. Incorrect accusations may affect honest vehicles through reputation, payment, or routing decisions; your evaluation must therefore examine false accusations as well as detection performance.

2 Scope and relationship to Assignment 2

Continue with the project track and core problem defined in Assignment 2. Refine the design as needed for implementation and explain material changes in the final report. Discuss a change of track or a substantial change of scope with the instructor before proceeding.

The dataset will be provided. You are not required to collect or generate the main dataset. Implement the diagnosis pipeline and its evaluation using the supplied evidence and documented labels or annotations, where available.

3 Expected outcome

Deliver a working research prototype that accepts evidence, runs the selected agentic workflow, produces a traceable diagnosis, and supports a reproducible comparison against simpler alternatives. A successful project may show either benefits or limitations of agentic diagnosis; claims must follow from measured results.

4 Projects Requirements

- Evidence handling. Load the supplied data, validate inputs, and handle missing evidence explicitly. Keep ground truth labels and any fields that reveal them out of agent inputs. Divide the data into training and testing.
- Agentic workflow. Implement identifiable agent roles with explicit objectives, permitted evidence, and structured outputs. Include at least one operational decision such as selecting an evidence tool, requesting further evidence, cross checking a claim, or revising a diagnosis. Explain how this decision affects the workflow.
- Evidence tools. Provide callable functions for inspecting or summarizing the available evidence. Restrict tools to documented inputs and distinguish observed facts from inferred explanations. A fixed sequence of prompts alone must not be presented as autonomous reasoning.
- Decision logic. Define the allowed diagnoses and how agent outputs become a final decision. At minimum, distinguish malicious behavior, benign degradation, and insufficient evidence. Map these outcomes to the supplied label scheme and explain any aggregation of detailed labels.
- Traceability. Save the diagnosis, supporting evidence references, a concise justification, uncertainty or confidence, and a record of agent and tool actions. Record only concise decision summaries; hidden model reasoning is not required.
- Operational reliability. Use bounded agent turns, tool calls, or retries. Handle invalid outputs, timeouts, and unavailable services without silently producing a confident diagnosis. Do not expose credentials in the submission.

A suggested output record contains case ID, predicted diagnosis, confidence or uncertainty, evidence references, brief justification, agent disagreement, execution status, and runtime. Numeric confidence is optional; if used, explain how it is obtained and do not assume it is a calibrated probability.

5 Track specific requirements

Track 1 Collaborative Diagnosis Agents

Implement all diagnosis agents with meaningfully different roles, evidence views, or reasoning responsibilities. Define what each agent can access and how the agents exchange findings. Implement a coordinator, voting rule, or other explicit aggregation mechanism that resolves disagreement or returns insufficient evidence.

Compare with other baselines like, collaboration with a single agent using the same underlying model and comparable evidence access. Include at least one ablation, such as removing a role or disabling cross checking, to examine what the collaboration contributes.

Track 2 Malicious Diagnosing Agents

Implement a diagnosis workflow in which at least one diagnosing agent can be made unreliable or malicious. Define the attacker's objective, access, and control. Implement at least two distinct behaviors, such as falsely accusing honest relays and suppressing supported malicious diagnoses. Separate attacks on diagnosing agents from malicious forwarding by vehicles.

Implement one defense, such as evidence validation, reliability weighting, or disagreement based review. Evaluate clean operation, attacked operation without the defense, and attacked operation with the defense under matched conditions. Vary attack intensity or the number of compromised agents across at least two nonzero settings.

6 Experimental design and evaluation

Specify evaluation cases, label mappings, baselines, metrics, and experimental settings before inspecting final test results. Use separate development and test partitions when tuning prompts, thresholds, or aggregation rules. Group related records by relay, session, scenario, or time where appropriate to reduce leakage, and document the split.

Use the supplied ground truth for scoring only. If labels or scenario coverage do not support a proposed experiment, agree on an alternative with the instructor and state the limitation. Do not invent missing ground truth or present synthetic evidence as observed data.

Required comparisons

- A simple nonagentic baseline, such as a documented threshold rule using available evidence. Tune any threshold on development data only.
- A single agent baseline, plus the relevant track comparisons specified in Section 5. Use matched evaluation cases and document differences in models, evidence access, or computation budgets.

Required measurements

Measure	Required reporting
Diagnosis quality	Confusion matrix and per class precision, recall, and F1; macro F1 over the labeled diagnosis classes.
False accusations	Benign cases predicted malicious divided by all ground truth benign cases. Report the numerator and denominator.
Uncertainty	Fraction of cases receiving a substantive diagnosis and fraction marked insufficient evidence. Report quality on decided cases alongside coverage.
Execution cost	Runtime per case and agent or tool calls. Report token use and estimated API cost where applicable.
Track evidence	Track 1: collaboration and ablation results. Track 2: attack impact, defense recovery, and defense effect on clean cases.

State how insufficient evidence is handled in each metric. Report failed executions separately and retain all test cases in the accounting.

Analysis and reproducibility

Where supported by the dataset, analyze results by malicious forwarding type, benign degradation cause, and evidence availability. For stochastic workflows, run at least three repetitions of the main comparisons and report variation; for deterministic workflows, state why repeated runs are unnecessary.

Record model identifiers and versions, prompts, configurations, software dependencies, seeds where supported, and evaluation commands. Include at least three examined cases: a correct malicious diagnosis, a benign case, and an incorrect or uncertain result. Relate each explanation to actual evidence and identify limitations.

8 Final submission

- **Source code and configuration**. Submit a repository link or archive containing the implementation, prompts, tool definitions, dependency specifications, and evaluation scripts. Include example outputs and machine readable results. Do not include API keys or redistribute restricted data.
- **README.** Explain installation, data placement, configuration, and exact commands for a sample diagnosis and the reported evaluation. State hardware, model or service requirements, estimated runtime, and known limitations. Provide saved traces so results can be reviewed without rerunning paid services.
- **Final report.** Write the report in a research paper format, with abstract, introduction, literature, proposed solution, evaluation, and conclusion. Include the problem and contribution; a concise updated literature comparison; changes from Assignment 2; architecture and workflow; dataset and experimental setup; results and case analysis; limitations; and conclusions.
- **Demonstration.** Prepare to present a **demonstration** of your implemented solution and usecases in addition to **presentation slides**. Show evidence input, agent and tool actions, the final diagnosis, and one uncertain or failure case.
- **Contribution statement**. Describe each member's implementation, experimentation, and writing contributions. Cite external code, datasets, papers, and models. Disclose AI assistance according to the course policy; each member must be able to explain the submitted work.

9 Proposed assessment rubric

Criterion	Weight
Working pipeline and implementation quality	30%
Agentic design and track specific mechanism	25%
Evaluation quality and analysis	25%
Report and demonstration	20%

Assessment emphasizes sound design, working implementation, fair comparisons, and honest analysis. A complex system or high accuracy alone does not establish a strong contribution. Unsupported claims and unexplained data leakage weaken the evidence for the project.